Shuyi (Ryan) Chen

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Contact

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Education

Carnegie Mellon University, Heinz College — Pittsburgh, PA
Ph.D. in Information Systems Management — 2024 – Summer 2027 (*expected*)
Advisors: Prof. Shixiang (Woody) Zhu and Prof. Ramteen Sioshansi
Master of Information Systems Management (Highest Distinction) — 2021 – 2023

The Chinese University of Hong Kong, Shenzhen — Shenzhen, China
B.B.A. (Hons.) in Global Supply Chain and Logistics Management; Minor in Statistics — 2017 – 2021
First Class Honours
Visiting Student, University of California, Berkeley — 2018

Research Interests

- Power-system resilience to wildfires and extreme weather
- Learning and optimization under uncertainty: decision-focused learning; robust and chance-constrained optimization; conformal prediction; generative models
- Application areas: grid hardening and undergrounding; sectionalization and Public Safety Power Shutoffs; outage management; energy storage operations

Publications

*Equal contribution.

Journal Articles

[J1] **Global-Decision-Focused Neural ODEs for Proactive Grid Resilience Management** [arXiv] [code]
Shuyi Chen, Ferdinando Fioretto, Feng Qiu, Shixiang Zhu.
IEEE Transactions on Smart Grid.

Refereed Conference Proceedings

[C1] **Learning Polyhedral Conformal Sets for Contextual Robust Optimization** [arXiv]
Shuyi Chen*, Wenbin Zhou*, Shixiang Zhu.
Conference on Neural Information Processing Systems (NeurIPS), 2026.

[C2] **Gen-DFL: Decision-Focused Generative Learning for Robust Decision Making** [arXiv]
Prince Zizhuang Wang, Shuyi Chen, Jinhao Liang, Ferdinando Fioretto, Shixiang Zhu.
International Conference on Learning Representations (ICLR), 2026.
IISE DAIS Track Best Paper Award.

[C3] **Counterfactual Fairness through Transforming Data Orthogonal to Bias** [arXiv] [code]
Shuyi Chen, Shixiang Zhu.
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), Research Track, 2025.

[C4] **Uncertainty-Aware Robust Learning on Noisy Graphs** [arXiv] [poster] [code]
Shuyi Chen, Kaize Ding, Shixiang Zhu.
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2025.
Short version at the NeurIPS 2023 Workshop on New Frontiers in Graph Learning.

[C5] **API-Assisted Code Generation for Question Answering on Varied Table Structures** [arXiv]
Yihan Cao*, Shuyi Chen*, Ryan Liu*, Zhiruo Wang, Daniel Fried.
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2023.

Working Papers

Under Revision

[W1] **Enhancing Electricity-System Resilience with Adaptive Robust Optimization and Conformal Uncertainty Characterization** [arXiv] [media]
Shuyi Chen, Shixiang Zhu, Ramteen Sioshansi.
Minor revision, IEEE Open Access Journal of Power and Energy.
INFORMS ENRE Best Student Paper Award.

In Preparation

[W2] **Large-Scale Resilience Planning for Wildfire-Prone Electricity Systems via Adaptive Robust Optimization** [arXiv] [poster] [demo]
Shuyi Chen, Shixiang Zhu, Ramteen Sioshansi.
To be submitted to European Journal of Operational Research.

[W3] **The Price of Looking Ahead: Decision-Focused Learning for Model Predictive Control in Energy Storage**
Shuyi Chen, Shuai Wang, Shixiang Zhu.
In preparation for conference submission. Earlier version: "Balancing Planning Myopia and Forecast Error: Decision-Focused Sequential Learning."
INFORMS DMDA Workshop Best Paper Finalist (Applied Track).

[W4] **Rare-Chance-Constrained Resilience Planning for Power Systems under Wildfires and Cascading Hazards**
Shuyi Chen, Shixiang Zhu, Ramteen Sioshansi.
In preparation for journal submission.

Honors & Awards

INFORMS DMDA Workshop Best Paper Finalist (Applied Track)	2026
IISE DAIS Track Best Paper Award	2026
INFORMS ENRE Best Student Paper Award	2025
KDD Student Travel Award; NSF Travel Grant (IMSI Workshop)	2025
INFORMS Workshop on Data Science Student Scholarship	2024
PwC Technology Impact Office / CMU Digital Transformation Center Research Fellowship	2023
CUHK Presidential Award (top 10 of 1,200 graduates)	2021

Teaching

Lecturer, *A/B Testing* (95-819), Heinz College, Carnegie Mellon University 2026
Teaching interests: optimization and operations research; machine learning and data analytics; A/B testing and causal inference; energy and sustainability analytics.

Presentations

Young Researchers Workshop, Cornell University 2026
Rare-chance-constrained resilience planning under wildfires and cascading hazards
HOPE Research Workshop, University of Notre Dame (invited talk) 2026
Large-scale resilience planning for wildfire-prone electricity systems
ACM SIGKDD Conference, Toronto 2025
Counterfactual fairness through transforming data orthogonal to bias
NSF/IMSI Workshop on UQ and ML for Complex Physical Systems, Chicago (poster) 2025
Electricity-system resilience with adaptive robust optimization and conformal uncertainty

INFORMS Workshops on Data Science and on Data Mining & Decision Analytics 2024
Counterfactual fairness; *decision-focused neural ODEs for grid resilience*

Industry Experience

Public Advocates Office, California Public Utilities Commission San Francisco, CA
Visiting Scholar Oct 2026 – present
Advisor: Dr. Iain Fisher
• Work with the office on wildfire resilience planning [W2] and on undergrounding decisions under rare, high-consequence hazards, motivated by California's Senate Bill 884 expedited undergrounding program.

Argonne National Laboratory Lemont, IL
Researcher 2025
• Developed decision-focused neural ODEs for proactive grid resilience, reducing decision regret by up to **75%** versus two-stage baselines [J1].
• Led the 2025 INFORMS DMDA Workshop Data Challenge on extreme-weather power-outage prediction, which drew **54 teams from 31 institutions**.

ByteDance Beijing, China
Data Scientist 2023 – 2024
• Built Prophet-based anomaly detection with statistical residual tests (**3% MAPE**), flagging three major business anomalies for timely mitigation.
• Applied causal inference to Local Services growth interventions, contributing to a **2% increase** in overall user activity.

Public Platform

- Public wildfire analytics platform (2026), which I built to help policymakers evaluate wildfire-mitigation policies: interactive analysis and download of California's wildfire, shutoff, and outage data; comparisons of planning strategies; and an AI assistant, built on large language models, that answers questions with cited data. [link]

Professional Service

Competition Co-Chair: INFORMS DMDA Workshop Data Challenge (2025); IISE ESD Energy Analytics Challenge (2026).
Organizing Committee: YinzOR, Speakers and Talks Committee (2026).
Session Chair: INFORMS Energy Systems session (2025).
Journal Reviewer: INFORMS Journal on Data Science; Journal of Agricultural, Biological, and Environmental Statistics.
Conference Reviewer: NeurIPS; ICML; ICLR; KDD; AISTATS; ICASSP; IEEE PES General Meeting (2023 – 2026).

References

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Principal Computational Scientist
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Ferdinando (Nando) Fioretto
Associate Professor, Computer Science
University of Virginia
fioretto@virginia.edu